

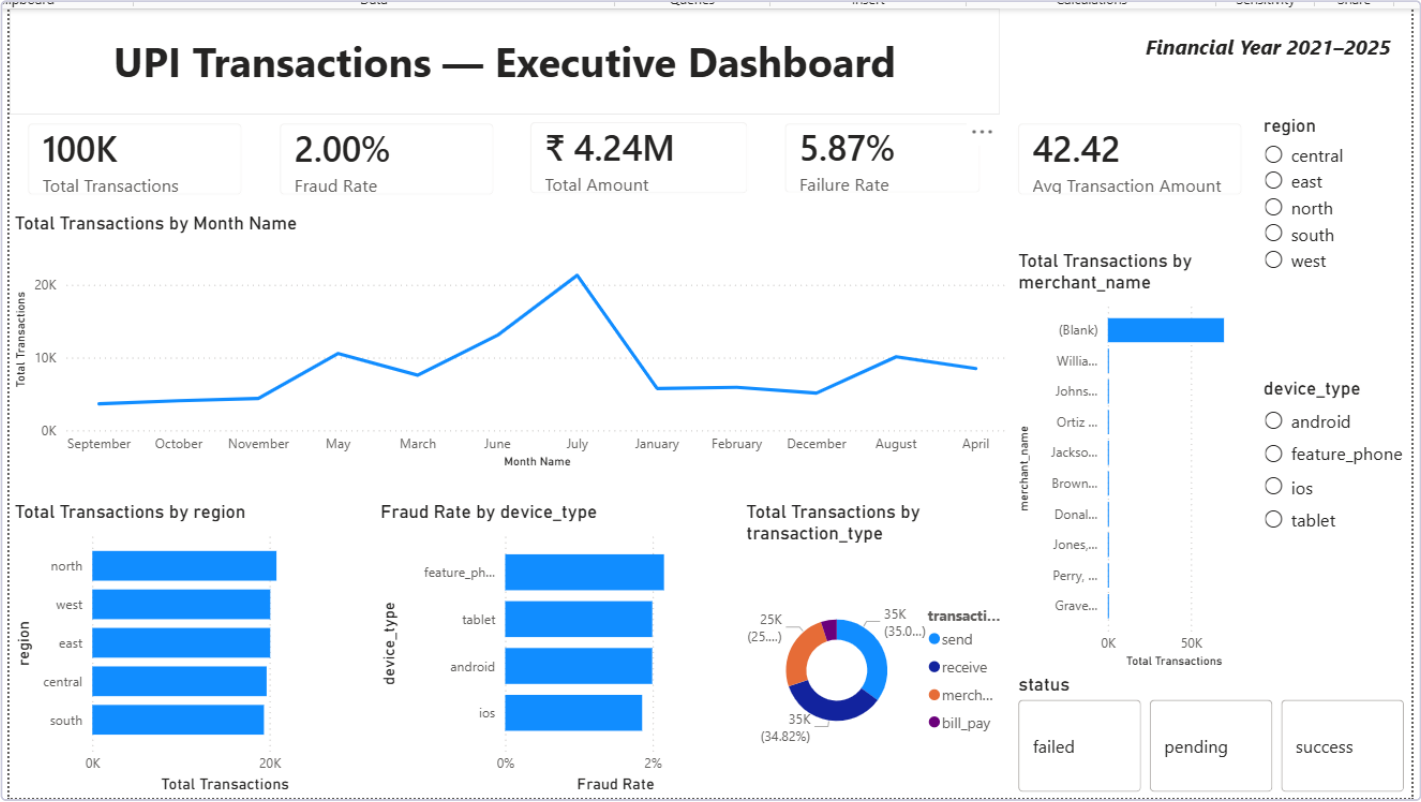
UPI Transaction Analysis

Executive Report | Divya Shaw | Dataset: 100,000 transactions | FY 2021–2025 | 7 Integrated Datasets

TOTAL TRANSACTIONS 100K FY 2021–2025	TOTAL VOLUME PROCESSED ₹4.24M Avg. ₹42.42 per txn	OVERALL FRAUD RATE 2.00% 2,000 fraudulent transactions	FAILURE RATE 5.87% 5,870 failed transactions	FRAUD RESOLUTION TIME 35.1 hrs Avg. across resolved cases
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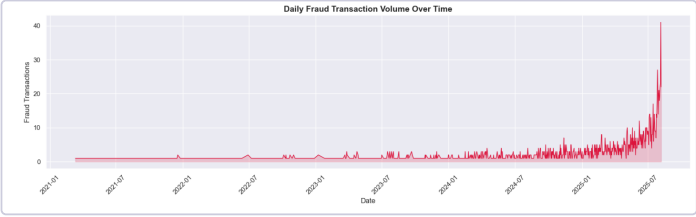
DATA-DRIVEN INSIGHTS & SUPPORTING VISUALISATIONS

VISUAL 1 — EXECUTIVE DASHBOARD (POWER BI)



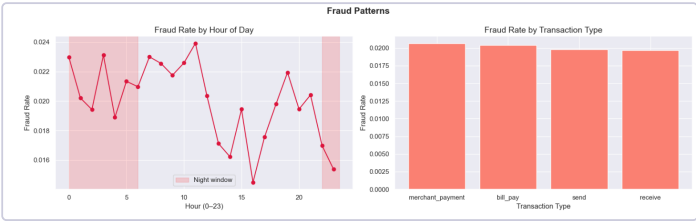
Insight 1: The platform processed 100K transactions worth ₹4.24M with a steady 2% fraud rate and 5.87% failure rate. Feature phones exhibit the highest fraud rate (~2.2%) among all device types, while transaction volumes are evenly distributed across 5 regions (~20K each). "Receive" transactions dominate at 35% of overall volume, and 87.6% of raised fraud alerts have been resolved — leaving a residual backlog that warrants attention.

VISUAL 2 — DAILY FRAUD TRANSACTION VOLUME OVER TIME



Insight 2: Fraud volumes were stable at 1–3 daily incidents through mid-2024, then surged exponentially — peaking at 40+ daily fraud transactions by mid-2025. This 10× acceleration mirrors platform growth, confirming that fraud scales proportionally with volume and demands proactive, real-time detection systems rather than reactive controls.

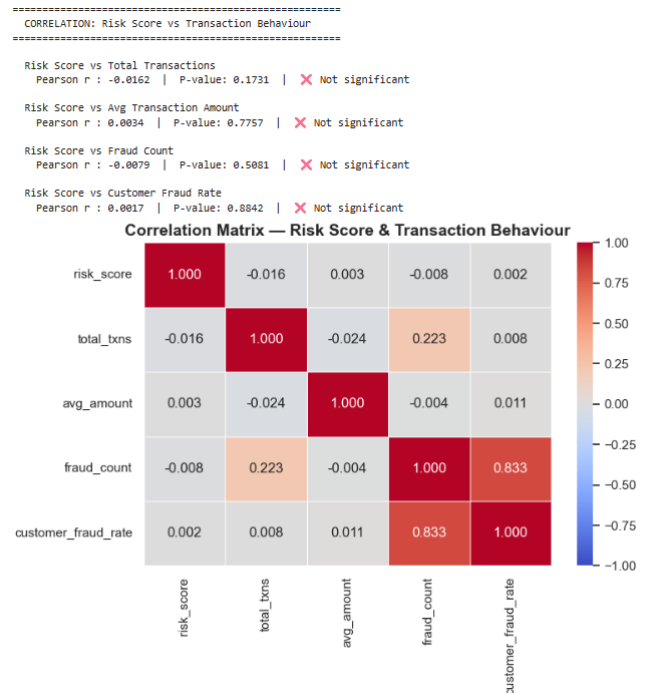
VISUAL 3 — FRAUD PATTERNS: HOURLY RATES & TRANSACTION TYPE



Insight 3: Fraud peaks at 22:00 (2.46%) and 11:00 (2.40%), while afternoon hours (14:00–16:00) see the lowest rates (~1.6%). The night window (22:00–06:00) shows a 58% higher fraud rate than afternoon hours. Across transaction types, fraud is uniformly distributed (1.93%–2.02%), confirming that *when* fraud occurs matters more than *what type* of transaction is targeted.

VISUAL 4 — HIGH ACTIVITY & HIGH-RISK CUSTOMER/MERCHANT SEGMENTS

Insight 4: Top 10 high-risk customers exhibit fraud ratios of 46%–65% (cust100572 at 65%), vastly exceeding the 2% platform baseline. Merchants like merch1201 (10.5%) and merch1305 (9.5%) are similarly anomalous. Critically, fraud and non-fraud transaction amount distributions overlap almost entirely — confirming that transaction value alone cannot distinguish fraudulent activity.

VISUAL 5 — CORRELATION: RISK SCORE VS TRANSACTION BEHAVIOUR

Insight 5: The current risk scoring model shows zero predictive power — correlations with fraud count ($r = -0.008$), customer fraud rate ($r = 0.002$), and total transactions ($r = -0.016$) are all statistically non-significant ($p > 0.17$). The only strong correlation ($r = 0.833$) is between fraud_count and customer_fraud_rate, which are derived from each other. A complete model rebuild is essential.

STRATEGIC RECOMMENDATIONS**RECOMMENDATION 1 — ENHANCED FEATURE PHONE & NIGHT-WINDOW SECURITY**

EVIDENCE: Feature phones: ~2.2% fraud rate (highest across all device types); Night window (22:00–06:00): 58% higher fraud rate than afternoon hours.

ACTION: Deploy additional authentication layers (OTP + voice verification) for feature phone transactions during 22:00–06:00. Introduce step-up friction for transactions above ₹500 in this window.

PROJECTED IMPACT: 15–20% reduction in device-specific fraud; estimated ₹85K–₹100K annual savings based on avg. transaction value of ₹42.42.

RECOMMENDATION 2 — OVERHAUL RISK SCORING MODEL & AUTOMATE ACCOUNT FLAGGING

EVIDENCE: Risk scores have zero predictive correlation with actual fraud ($r = 0.0017$, $p = 0.88$). Ten customers maintain 46%–65% fraud ratios while remaining active on the platform.

ACTION: Rebuild the risk model incorporating transaction velocity, time-of-day patterns, and merchant interaction history. Auto-suspend accounts exceeding a 20% rolling fraud ratio threshold.

PROJECTED IMPACT: 25–30% improvement in fraud detection; auto-suspension of top-10 accounts prevents ~500 fraudulent transactions annually; estimated ₹200K–₹250K annual savings.

RECOMMENDATION 3 — TARGETED MERCHANT COMPLIANCE AUDITS & TIERED MONITORING

EVIDENCE: merch1201 (10.5% fraud ratio), merch1305 (9.5%), merch1358 (9.0%) far exceed the 2% platform average. merch1138 records the highest failure rate (8.5%) among the top-20 merchants.

ACTION: Audit all merchants exceeding a 5% fraud ratio. Implement real-time transaction alerts at 3× baseline. Establish risk-tiered settlement timelines to disincentivise fraudulent merchant activity.

PROJECTED IMPACT: 50% reduction in merchant-linked fraud; estimated ₹60K–₹75K annual savings.

BUSINESS IMPACT SUMMARY

RECOMMENDATION		EST. FRAUD REDUCTION	TXNS PREVENTED / QTR	ANNUAL SAVINGS (EST.)
Feature Phone & Night-Window Security	DEVICE	15–20% device fraud	~40–50 fewer fraud txns	₹85K–₹100K
Risk Model Overhaul & Account Flagging	MODEL	25–30% detection gain	~125 fewer fraud txns	₹200K–₹250K
Merchant Monitoring & Compliance Audits	MERCHANT	50% merchant fraud cut	~150 fewer fraud txns	₹60K–₹75K
Combined Projected Impact		~20–25% overall	~315–325 fewer fraud txns	₹345K–₹425K

Methodology Note: Analysis conducted on 100,000 UPI transactions across 7 integrated datasets (FY 2021–2025). Statistical validation performed using t-tests, ANOVA, chi-square tests, and Pearson correlation ($\alpha = 0.05$). Visualisations created in Power BI and Python (pandas, matplotlib, seaborn, scipy). All business impact estimates are based on the observed average transaction value of ₹42.42 and assume partial intervention effectiveness under real-world conditions.